UNIT 1 · TOPIC 1.3

Best Practices for Public Networks

Quiz, Answer Key

Teacher answer key, 18 questions, 1 point each. The correct answer is highlighted; the explanation and distractor notes follow.

1. Which term describes a rogue wireless access point that broadcasts the same network name (SSID) as a legitimate nearby network in order to trick users into connecting to it?

- ● **A. Evil twin attack**
- ○ B. Deauthentication attack
- ○ C. ARP poisoning attack
- ○ D. Replay attack

Correct answer: A

Why: An evil twin is a fraudulent access point configured to broadcast an SSID identical to a trusted network. Users who connect unknowingly expose their traffic to the adversary. Deauthentication attacks force disconnection from a legitimate AP; ARP poisoning corrupts local address-resolution tables to redirect traffic at Layer 2; replay attacks retransmit captured packets to impersonate a prior session. None of these involve cloning a network name to lure users.

*Why the others are wrong: **B**, Students confuse deauthentication with impersonation because both can precede a credential-theft attack; deauthentication is the kick-off mechanism, not the rogue AP itself. **C**, ARP poisoning is a plausible distractor because it also redirects traffic, but it operates at Layer 2 within a network, not by mimicking a network's identity. **D**, Replay attacks involve retransmitting captured credentials or tokens, not creating a fake access point, students who focus on "capturing" data may conflate the two.*

2. A person sets up an unsecured wireless hotspot in a coffee shop without authorization, intending to intercept login credentials from nearby customers for personal financial gain. Which category of adversary does this most accurately describe?

- ○ A. Nation-state threat actor
- ● **B. Opportunistic cybercriminal**
- ○ C. Hacktivist
- ○ D. Insider threat

Correct answer: B

Why: Opportunistic cybercriminals exploit accessible, low-effort targets for personal financial gain. Setting up a rogue hotspot in a public venue requires minimal resources and targets unaware bystanders, a hallmark of opportunistic attack behavior. Nation-state actors and hacktivists are driven by geopolitical or ideological motives and typically pursue higher-value targets through more sophisticated means. Insider threats originate from within an organization that grants them privileged access.

*Why the others are wrong: **A**, Students may overestimate the sophistication of a rogue-AP attack and associate it with state-level resources; the attack is actually within reach of low-skill individuals. **C**, Hacktivists are motivated by ideology rather than profit, a common confusion when students see "attacker" without thinking about motive. **D**, Insider threats require a pre-existing relationship with the target organization; a random coffee-shop attacker has no such position.*

3. In which type of wireless attack does an adversary secretly position themselves between a user and a legitimate access point, intercepting and potentially altering communications without either party's knowledge?

- ○ A. Brute force attack
- ○ B. SQL injection
- ○ C. Denial of service (DoS) attack
- ● **D. Man-in-the-middle (MitM) attack**

Correct answer: D

Why: In a man-in-the-middle attack, the adversary relays communications between two parties while invisibly reading or modifying them. On public Wi-Fi, this is commonly achieved by controlling the access point or using ARP poisoning. Brute force attacks systematically try passwords; SQL injection targets database input fields; DoS attacks overwhelm a service to deny availability. None of these involve secretly intercepting traffic in transit.

*Why the others are wrong: **A**, Students who associate "attack on Wi-Fi" with password cracking may select brute force; the distinction is that brute force targets credentials at login, not traffic in transit. **B**, SQL injection is sometimes introduced alongside network topics, leading students to confuse web-application and network-layer attacks. **C**, DoS is a high-profile attack type that students may default to; the key difference is that DoS disrupts availability rather than harvesting data covertly.*

4. A student at a transit hub connects to a network labelled "Free_Transit_WiFi" and is redirected to a captive portal that requests her social media credentials. She enters them and gains internet access, but later discovers her social media password has been changed by an unknown party. Which combination of wireless attack techniques does this scenario most precisely illustrate?

- **● A. Evil twin access point combined with credential harvesting via a fake captive portal**
- ○ B. Deauthentication attack followed by a replay attack
- ○ C. ARP poisoning followed by session hijacking at the router
- ○ D. Bluejacking combined with bluesnarfing over Bluetooth

Correct answer: A

Why: The adversary cloned the legitimate network's SSID to create an evil twin, then deployed a fake captive portal to harvest the victim's social media credentials, a compound attack documented in NIST SP 800-153. Option B involves forced disconnection and retransmission of captured tokens, not credential harvesting via a portal. Option C operates at Layer 2 within an existing network and does not involve SSID spoofing. Option D involves Bluetooth proximity attacks unrelated to Wi-Fi.

*Why the others are wrong: **B**, Students who know that deauthentication is sometimes used to force users off a legitimate AP may select this, confusing the setup step with the full attack chain. **C**, ARP poisoning is a plausible distractor because it also redirects traffic; students may not distinguish it from SSID-level impersonation. **D**, Bluetooth attacks are a common exam distractor; the presence of "wireless" in the question may cause students to over-include Bluetooth as a possible vector.*

5. A user submits their online banking password over an unencrypted HTTP connection while connected to a public Wi-Fi network. Which statement most accurately describes why this creates a cybersecurity risk?

- ○ A. The banking server's firewall will block the unencrypted request before it reaches the server
- ○ B. WPA2 encryption on the access point protects the credentials from any third-party interception
- **● C. The credentials are transmitted as readable plaintext and can be captured by any device monitoring the same network segment**
- ○ D. The browser automatically hashes the password before sending it over HTTP

Correct answer: C

Why: HTTP transmits data without encryption at the application layer. On a shared public network, any device performing passive packet capture can read credentials in plaintext. WPA2 encrypts the wireless channel between the device and the access point but does not protect application-layer data once decrypted at the AP, or at all if the AP itself is adversary-controlled. Firewalls filter traffic by port and protocol rules, not by content readability. Standard browsers do not hash passwords before HTTP submission.

*Why the others are wrong: **A**, Students who conflate server-side security with in-transit protection may believe a firewall substitutes for encryption. **B**, WPA2 is a common misconception here, students often assume that because Wi-Fi has a password, all data on it is secure from eavesdropping. **D**, This distractor targets students who confuse client-side hashing in specific authentication protocols with standard browser HTTP behavior.*

6. An adversary monitors unencrypted traffic at an airport lounge, collects plaintext email addresses and passwords, and tests those credentials against a banking website. Many stolen passwords work on the banking

site because users reused them across services. How does this attack sequence most accurately illustrate the risk generated by credential theft on public networks?

○ A. It compromises network infrastructure integrity but does not affect individual account confidentiality

● B. It enables credential stuffing across multiple services, creating account-takeover risk that extends far beyond the original breach point

○ C. It primarily affects the availability of the airport network rather than the confidentiality of user data

○ D. It introduces malware onto victims' devices via the wireless interface, enabling persistent unauthorized access

Correct answer: B

Why: Credential stuffing uses stolen username-password pairs from one service to gain unauthorized access to others, exploiting the widespread practice of password reuse. The risk therefore multiplies across every service sharing those credentials, not just the one where interception occurred. This attack does not alter network infrastructure, degrade network availability, or implant malware, it is a passive interception and reuse attack targeting confidentiality and authentication integrity.

*Why the others are wrong: **A**, Students may confuse "network attack" with "infrastructure attack" and miss that the primary victim is the individual user's account portfolio. **C**, DoS attacks target availability; passive sniffing does not. Students may select C if they conflate any public-network attack with service disruption. **D**, Malware implantation is a separate attack class; passive sniffing harvests data in transit but does not deliver executable code to the victim's device.*

7. In Scenario 1C, the victim joins "Guest Wi-Fi" instead of "Sunshine Coffee Wi-Fi," enters streaming platform credentials into a fake captive portal, and is later locked out of her music account. Which sequence of attack stages most completely and accurately characterizes what occurred?

○ A. Reconnaissance → Exploitation → Exfiltration → Privilege escalation

● B. Network spoofing → Credential harvesting → Credential stuffing → Account takeover

○ C. Phishing email → Malware installation → Lateral movement → Persistence

○ D. ARP poisoning → Session hijacking → Data exfiltration → Denial of service

Correct answer: B

Why: The attack unfolded in four discrete stages: the adversary cloned the coffee shop's SSID to create "Guest Wi-Fi" (network spoofing); deployed a fake captive portal to collect OAuth credentials (credential harvesting); tested those credentials against linked services including the music platform (credential stuffing); and changed the password to lock out the legitimate user (account takeover). Option A applies generic kill-chain vocabulary without mapping to the specific Wi-Fi mechanisms present. Option C describes a phishing-to-malware chain that did not occur. Option D involves ARP-layer manipulation, which was unnecessary here because the adversary controlled the AP entirely.

*Why the others are wrong: **A**, Students familiar with MITRE ATT&CK may apply its abstract stages correctly in spirit but fail to identify the specific technical mechanisms, which is what this question requires. **C**, Phishing-to-malware chains are heavily emphasized in cybersecurity instruction; students may default to this pattern even when the scenario clearly involves a network-layer attack. **D**, ARP poisoning and session hijacking are legitimate Wi-Fi attack techniques, making this a high-quality distractor for students who understand Layer 2 attacks but misapply them here.*

8. Evaluate the following claim: "Setting up an evil twin access point requires specialized hardware and advanced networking expertise, making it an unlikely threat from opportunistic adversaries." Which response most accurately critiques this assertion?

○ A. The claim is correct; evil twin attacks require custom firmware unavailable to non-expert attackers

○ B. The claim is partially correct; initial setup is simple, but avoiding detection requires advanced skills

● C. The claim is incorrect; commodity tools and consumer-grade hardware reduce the technical barrier for evil twin attacks to a level accessible to low-skill adversaries

○ D. The claim is incorrect; evil twin attacks are only feasible for nation-state actors because of the cost of the required equipment

Correct answer: C

Why: Dedicated Wi-Fi auditing devices, widely available open-source software (such as hostapd and airbase-ng), and even smartphone hotspot features collectively make evil twin attacks accessible without specialized expertise, often for under $50 in hardware. The attack does not require custom firmware or state-level resources. Option B conflates operational security (avoiding detection) with execution difficulty, a real but distinct concern. Options A and D dramatically overestimate the barrier to entry, which is precisely the misconception this question surfaces and corrects.

*Why the others are wrong: **A**, Students with limited exposure to attacker toolkits may assume that network-layer attacks are inherently complex; this distractor tests that assumption directly. **B**, This is the most dangerous distractor, it contains a true statement (detection avoidance is harder) embedded in a partially incorrect framing, testing whether students can distinguish execution complexity from operational complexity. **D**, Nation-state framing is a common misconception driven by high-profile breach media coverage; students may overgeneralize it to all network attacks.*

9. A security analyst reviews an incident where hotel guests joined a rogue access point, authenticated through a fake captive portal, and had multiple linked accounts subsequently compromised. Which combination of threat characteristics most completely describes the attack surface that was exploited?

○ A. Local access vector · medium complexity · software vulnerability exploitation · persistence via installed malware

○ B. Network access vector · high attack complexity · cryptographic weakness in WPA2 · single-service account impact

● C. Physical proximity vector · low attack complexity · credential-based authentication weakness · multi-service credential reuse

○ D. Adjacent access vector · high complexity · zero-day exploit · privilege escalation to administrator

Correct answer: C

Why: The attack required the adversary to be physically present in or near the hotel (physical proximity vector); it exploited users' tendency to join open SSIDs and enter credentials into any portal presented (low complexity, no cryptographic breaking required); the attack surface was authentication, specifically users surrendering credentials to an unverified portal; and the cascade across accounts depended entirely on credential reuse across linked services. No cryptographic weakness, malware, software vulnerability, zero-day exploit, or privilege escalation was involved, eliminating options B, C, and D.

*Why the others are wrong: **A**, Malware appears frequently in incident reports; students may assume that compromise requires code execution rather than recognizing that credential reuse alone is sufficient. **B**, Students who equate "wireless attack" with "cryptographic attack" may select this, confusing WPA2 cracking (which did not occur) with the social-engineering-adjacent nature of the real attack. **D**, Zero-day exploits and privilege escalation appear in high-profile breach reporting; students may over-attribute complexity to attacks that are actually elementary in execution.*

10. Which of the following security controls most directly protects data transmitted over a public Wi-Fi network by creating an encrypted tunnel for all outbound traffic?

○ A. A strong, unique account password

○ B. Multi-factor authentication (MFA)

● C. A Virtual Private Network (VPN)

○ D. A device screen lock PIN

Correct answer: C

Why: A VPN encrypts all traffic between the user's device and a trusted exit point before it traverses the public network, protecting data in transit regardless of the underlying network's security posture. Strong passwords protect accounts at login but do not encrypt network traffic. MFA adds a second authentication factor, reducing account-takeover risk, but does not encrypt the communication channel. A screen lock PIN prevents physical device access and has no effect on data transmitted over a network.

*Why the others are wrong: **A**, Students who conflate account security with network security may select a strong password, not recognizing that password strength is irrelevant to in-transit data interception. **B**, MFA is frequently taught alongside VPN as a best practice; students may select it because it is a genuine security control, without distinguishing its scope (authentication) from channel encryption. **D**, Screen locks are visible and commonly discussed; students may include them as a reflexive response to any security question about devices.*

11. What security property does HTTPS provide that plain HTTP does not, making HTTPS preferable when transmitting sensitive information on any network?

○ A. It hides the user's IP address from the destination web server

○ B. It prevents the web server from creating logs of user activity

○ C. It prevents third parties from identifying the domain names the user visits

● D. It encrypts data transmitted between the user's browser and the web server

Correct answer: D

Why: HTTPS uses TLS (Transport Layer Security) to encrypt the data exchanged between the client browser and the web server, making intercepted traffic unreadable to eavesdroppers. It does not mask the client's IP address (that requires a proxy or VPN), does not hide domain names from third parties (the domain remains visible via DNS queries and the TLS Server Name Indication field), and does not prevent server-side logging, the server decrypts the content it receives and may log it freely.

*Why the others are wrong: **A**, VPNs and HTTPS both protect privacy but in different ways; students often merge the two and attribute IP-hiding to HTTPS. **B**, Students who conflate privacy with encryption may believe that encrypting traffic also prevents the server from seeing what it receives, missing that the server is one endpoint of the encrypted session. **C**, Domain-name visibility is a genuine limitation of HTTPS that many students are unaware of, making this an effective distractor for testing depth of understanding.*

12. Which of the following actions best reduces the risk of data interception when a person must use a public Wi-Fi network?

● A. Using HTTPS websites and activating a VPN for all connections

○ B. Disabling Bluetooth and connecting to any available open network

○ C. Selecting the network with the strongest signal to ensure the fastest connection

○ D. Logging out of sensitive accounts only after returning home from the public location

Correct answer: A

Why: Combining HTTPS (application-layer encryption per connection) with a VPN (full-tunnel network encryption) addresses both scenarios where individual connections are unencrypted and scenarios where the access point itself may be adversary-controlled. Signal strength indicates proximity to an access point, not trustworthiness. Disabling Bluetooth removes a separate attack surface but does not protect Wi-Fi traffic. Delayed logout leaves active sessions exposed to session hijacking for the entire duration of the public network connection.

*Why the others are wrong: **B**, Disabling Bluetooth is a genuine security hygiene step, making this distractor plausible; students may not recognize that it addresses a different attack surface from Wi-Fi interception. **C**, Strong signal is often associated with better performance; students may equate signal quality with security without understanding that rogue APs can deliberately boost signal to attract users. **D**, Students may believe logout timing is the primary session protection; this tests understanding that session exposure happens during the session, not only after it ends.*

13. A user connects to an untrusted public Wi-Fi access point and activates a VPN before accessing any websites. How does the VPN specifically mitigate the risk of data interception on this network?

○ A. The VPN authenticates the access point's certificate, preventing users from connecting to rogue APs

○ B. The VPN replaces WPA2 wireless encryption, making the underlying access point's security configuration irrelevant

○ C. The VPN blocks all unsolicited inbound connections, functioning as a host-based firewall on the public network

● D. The VPN encrypts all outbound traffic before it leaves the device, so any data intercepted on the public network remains unreadable to the adversary

Correct answer: D

Why: A VPN encapsulates and encrypts the user's traffic at the device level before transmission. An adversary monitoring the public network segment sees only encrypted ciphertext, not the content of the communications. VPNs do not authenticate access points, that function belongs to 802.1X/EAP-based protocols. VPNs are not host-based firewalls

(though some VPN clients include firewall features, that is not the core mitigation mechanism). VPNs supplement rather than replace wireless channel encryption; both layers operate simultaneously per NIST SP 800-77 guidance.

*Why the others are wrong: **A**, Students who know that VPNs involve certificate-based authentication may incorrectly attribute AP authentication to the VPN rather than to the 802.1X protocol stack. **B**, Students who do not clearly separate VPN (application/network layer) from WPA2 (link layer) may assume the two are redundant, a common conceptual gap. **C**, Some VPN clients do include firewall features, making this a plausible distractor; the question tests whether students can identify the primary encryption function versus ancillary features.*

14. A student needs to submit a graded assignment through her school's online portal while sitting in a public library. She wants the combination of controls that best protects her credentials and submission. Which set of actions provides the most complete risk mitigation for this situation?

● A. Confirm the correct library Wi-Fi name with a staff member, use only HTTPS connections, and activate a VPN before logging in

○ B. Connect to any available network and rely on the school portal's password complexity requirements

○ C. Select the network with the most connected users and enable file sharing for faster upload speeds

○ D. Use the library's Ethernet port if a VPN is unavailable, without requiring HTTPS

Correct answer: A

Why: A layered approach addresses multiple risk points: verifying the SSID with library staff prevents connection to an evil twin; HTTPS encrypts the application-layer communication with the school portal; and a VPN encrypts the full network tunnel, protecting against adversaries who may control the access point. This aligns with the defense-in-depth principle recommended by NIST SP 800-153. Option B relies solely on server-side password policy, which does not protect credentials in transit. Option C conflates network popularity with security. Option D partially mitigates wireless eavesdropping but still transmits unencrypted application-layer data without HTTPS.

*Why the others are wrong: **B**, Password complexity is frequently taught as a primary control; students may overestimate its relevance to in-transit data protection. **C**, A busy network may feel safer due to perceived oversight; this distractor targets social-intuition misconceptions about public network security. **D**, Ethernet is a partial mitigation that eliminates wireless eavesdropping; this distractor tests whether students understand that wired connections do not guarantee application-layer protection without HTTPS.*

15. A user's smartphone is configured to automatically rejoin any Wi-Fi network it has previously connected to. Why is disabling this auto-connect feature considered a security control for users who frequently use public networks?

○ A. It reduces battery consumption, which prevents the device's location from being tracked via signal-strength analysis

● B. It prevents the device from silently joining a rogue access point that broadcasts an SSID matching a previously trusted network

○ C. It stops adversaries from sending deauthentication frames that force the device off its current legitimate connection

○ D. It encrypts the list of stored Wi-Fi network names held in the device's configuration

Correct answer: B

Why: Auto-connect uses the device's stored SSID list to rejoin networks without user confirmation. An adversary who knows common public SSIDs can broadcast those names to silently pull reconnecting devices onto a rogue AP. Disabling auto-connect forces the user to manually select and verify a network before joining, breaking this passive attack path. Battery usage and location tracking are unrelated security properties in this context. Deauthentication attacks operate at the 802.11 management frame level and are not addressed by auto-connect settings. Auto-connect configuration does not affect the encryption of stored credentials, which is managed by the device keystore independently.

*Why the others are wrong: **A**, Battery drain and location privacy are tangentially associated with Wi-Fi scanning; students reasoning from "less scanning = less exposure" may select this. **C**, Deauthentication attacks and auto-connect are both Wi-Fi management concepts; students may conflate them because both relate to how devices join and leave networks. **D**, Students who know that stored credentials should be protected may assume a setting related to stored networks also encrypts them, conflating two separate security properties.*

16. In Scenario 1C, the victim could have applied security controls at three stages: before joining the network, while connected, and after the session. Which layered sequence of controls would have been most effective in preventing the credential theft?

○ A. Before: connect to the highest-signal network; During: use HTTP for faster speeds; After: change passwords when the service prompts

○ B. Before: disable VPN to allow captive portal access; During: rely on MFA alone; After: review browser history for suspicious sites

○ C. Before: enable auto-connect for convenience; During: use OAuth login for speed; After: run an antivirus scan

● D. Before: verify the correct SSID with coffee shop staff; During: decline OAuth login on any captive portal and activate a VPN; After: monitor linked accounts for unauthorized changes

Correct answer: D

Why: A layered control sequence addresses each stage of the attack chain: confirming the SSID before connecting prevents joining the evil twin; refusing OAuth login on an unverified captive portal removes the credential-harvesting opportunity (legitimate captive portals do not require platform login credentials, the request itself is a red flag); a VPN encrypts traffic even if the AP is adversary-controlled; and post-session account monitoring enables rapid response if any compromise did occur. Options A and C preserve or introduce the specific behaviors the adversary exploited. Option B disables the VPN at precisely the most critical moment, creating the vulnerability the scenario illustrates rather than preventing it.

*Why the others are wrong: **A**, Students may default to "faster = better" reasoning and not recognize that signal strength and HTTP have no bearing on protection against evil twin attacks. **B**, This distractor describes a partially correct policy (VPN required generally, MFA used) but introduces the specific exception that created the real-world vulnerability, testing students' ability to identify policy gaps, not just list controls. **C**, Convenience-focused behaviors (auto-connect, OAuth) are described as dangerous in the scenario; this distractor tests whether students can identify harmful defaults as such.*

17. A network policy analyst evaluates a proposal to replace open captive portals at public venues with 802.1X port-based access control. Which argument most accurately supports this change as a risk mitigation measure against the attack type illustrated in Scenario 1C?

○ A. 802.1X requires users to create unique venue-specific credentials, increasing password diversity across services

○ B. 802.1X encrypts the SSID broadcast itself, preventing adversaries from cloning the network name

● C. 802.1X eliminates browser-based captive portals, removing the credential-harvesting attack surface they represent

○ D. 802.1X mandates multi-factor authentication, making any account takeover impossible once deployed

Correct answer: C

Why: The attack in Scenario 1C succeeded because the victim was presented with a browser-based captive portal soliciting platform credentials, an attack surface that 802.1X eliminates. 802.1X authenticates users at the network layer via EAP and a RADIUS server before any browser is invoked, removing the possibility of a fake web-based portal. 802.1X does not encrypt SSID broadcasts (beacon frames remain unencrypted regardless of authentication method). It does not inherently mandate MFA, that depends on RADIUS server configuration. It does not generate unique per-venue passwords; authentication uses existing enterprise credentials or certificates.

*Why the others are wrong: **A**, Students may conflate "enterprise authentication" with "unique passwords per venue," missing that 802.1X authenticates against a central directory rather than creating new credentials. **B**, SSID encryption is a genuine limitation of current 802.11; students may assume 802.1X addresses it by association with "security," without understanding what 802.1X actually controls. **D**, MFA is often presented as near-universal best practice; students may attribute it to 802.1X without understanding that 802.1X is an authentication framework, not a specific authentication method.*

18. A company policy requires all employees to use a VPN on public networks, but includes an exception allowing them to temporarily disable the VPN to complete captive portal login. A security auditor flags this exception as a significant vulnerability. Which explanation most precisely justifies the auditor's concern?

○ A. The VPN actively inspects captive portal TLS certificates, and disabling it prevents certificate validation, increasing man-in-the-middle exposure at the application layer

○ B. Disabling the VPN prevents automatic reconnection after portal authentication, leaving all subsequent session traffic unencrypted indefinitely

○ C. Disabling the VPN deactivates WPA2 on the device, leaving the wireless channel permanently unencrypted for the rest of the session

● D. Disabling the VPN to authenticate through a captive portal creates an unprotected window during which pre-tunnel traffic, including captive portal credentials, can be intercepted by an adversary controlling the access point

Correct answer: D

Why: Before the VPN tunnel is established, the device communicates over the network in the clear. A fake captive portal can harvest credentials during this unprotected pre-tunnel window, precisely the mechanism in Scenario 1C. The policy exception essentially permits the exact attack the VPN requirement was designed to prevent. The VPN does not inspect captive portal certificates on behalf of the user; TLS certificate validation is performed by the browser's trust store independently of VPN state. Disabling a VPN does not affect WPA2, which operates at the link layer independently. Most enterprise VPN clients are configured to reconnect automatically after captive portal authentication, making option B factually incorrect for standard deployments.

*Why the others are wrong: **A**, TLS inspection is a real enterprise VPN/proxy capability; students who know about deep-packet inspection in corporate environments may incorrectly attribute this to standard consumer VPN clients. **B**, This is the most plausible wrong answer: it correctly identifies that VPN disconnection could leave traffic exposed, but misattributes the cause to reconnection failure rather than the pre-tunnel window, a subtle distinction that rewards careful reading. **C**, Students who do not clearly separate VPN (network/application layer) from WPA2 (link layer) may believe disabling one affects the other, a common network-stack conceptual gap.*